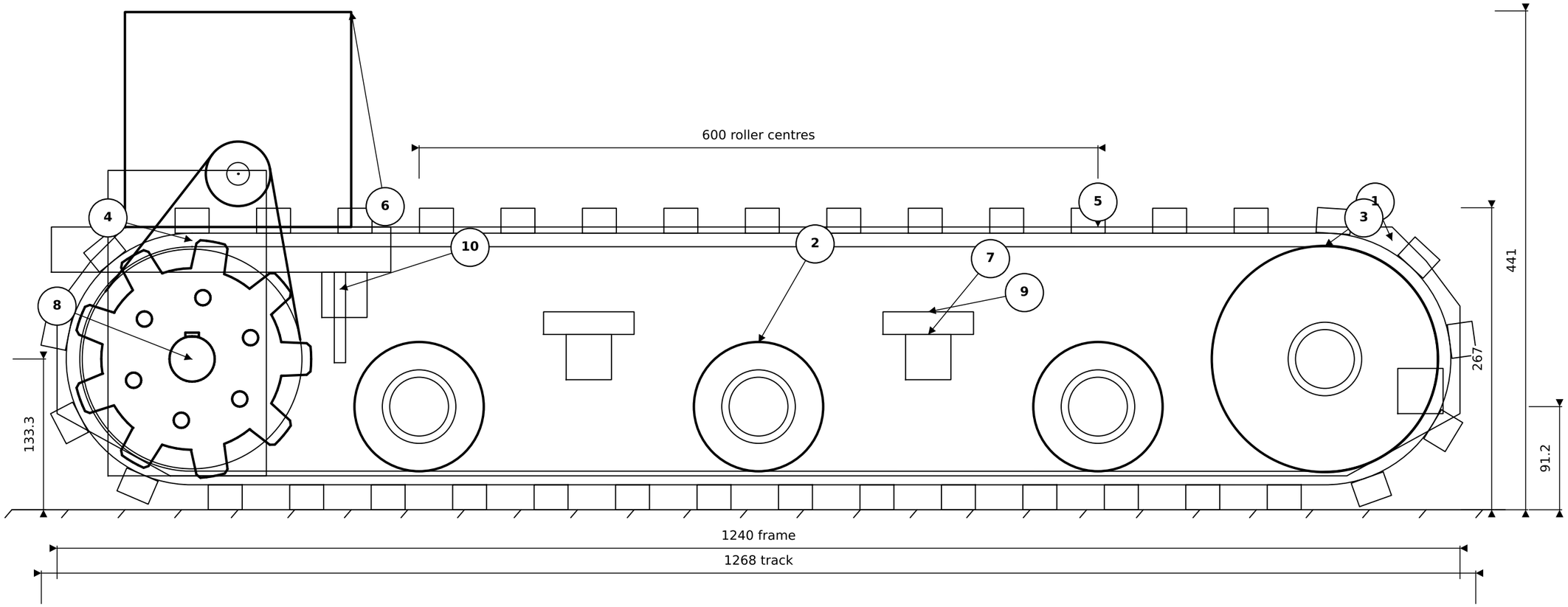
## SIDE ELEVATION (scale 1:4)




**NOTES**

1. Near side plate removed for clarity; running gear shown true.
2. Datum: ground at lug tips z=0, X travel, Z up. Dimensions mm.
3. Track loop: 37 links x 72 pitch, R105.26, C1001.3 (closure 0).
4. Chain plane y=+/-140; 48T/14T 428, 60 links, C~170.
5. Overall W826 over UCFL208 flanges; plates 726; track 700.
6. Motor body y 4..134 (inboard); chain runs at y=140, outboard of motor.

**FLAIL MOWER 500 / UC-V2**

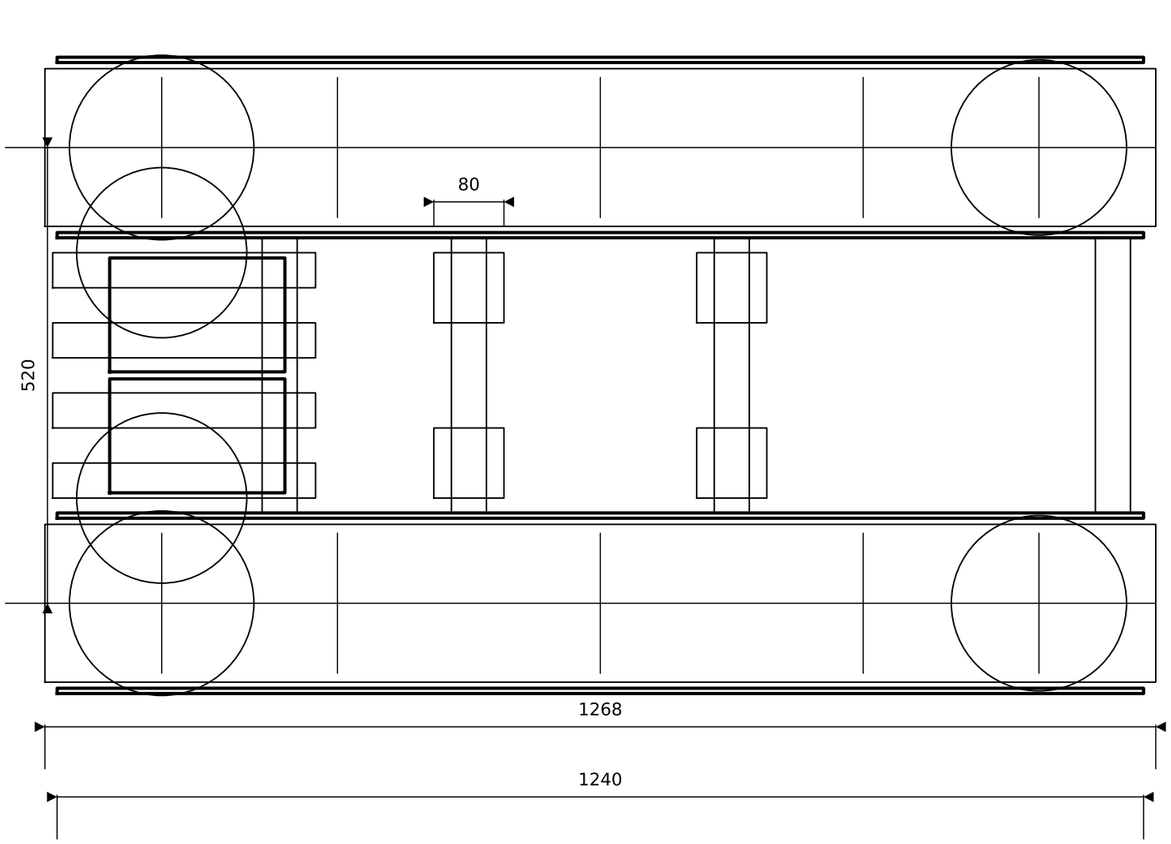
**GENERAL ARRANGEMENT - SIDE ELEVATION**

DWG No.
**UCV2-GA-001**
REV A

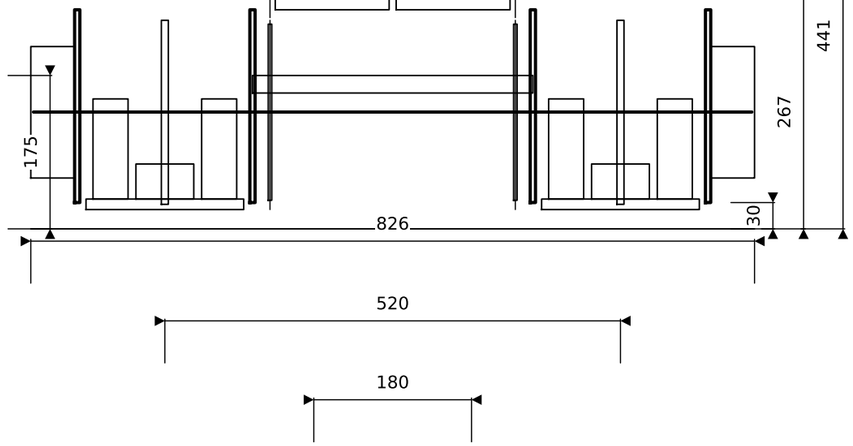
| MATERIAL | QTY / MASS |
|---|---|
| - | 1 / 199.9 kg |

| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:4 | A3 | 1 / 14 | 2026-10-29 | OC |

## PLAN (1:7)




## END VIEW (1:7)




**NOTES**

1. Track centres 520; plate gauge 200; overall 826 over UCFL.
2. Deck pads: x +/-150, y +/-100, top z175 (4x M12).
3. Motors 2x 1.2 kW on rails at z250..441, pinions at y +/-140.
4. Lift lugs at x=-370, y +/-100.

**FLAIL MOWER 500 / UC-V2**

# GENERAL ARRANGEMENT - PLAN & END VIEW

DWG No.

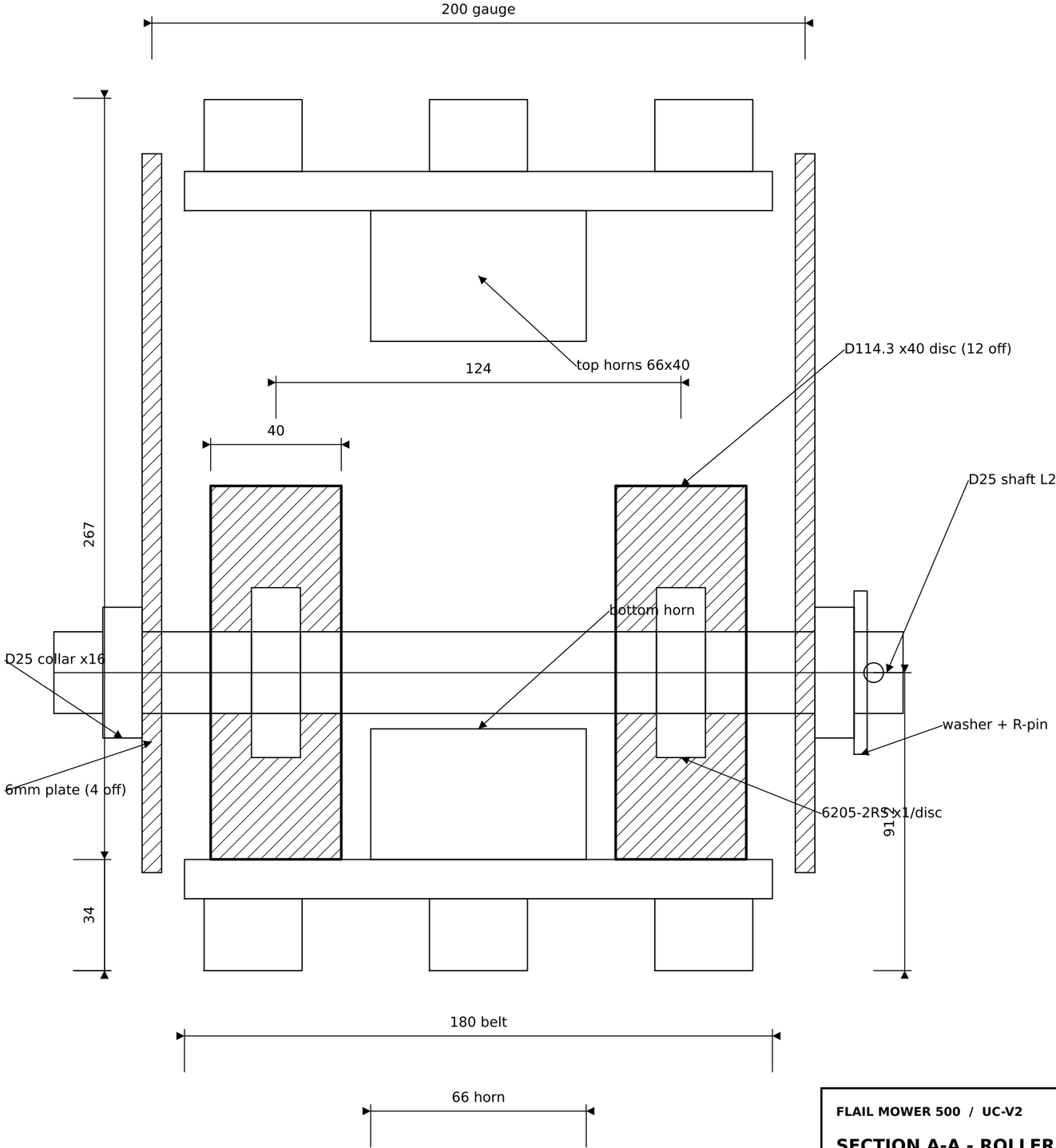
**UCV2-GA-002**

REV A

| MATERIAL | QTY / MASS |
|---|---|
| - | 1 / 199.9 kg |

| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:7 | A3 | 2 / 14 | 2026-10-29 | OC |

## SECTION A-A (1:1.5)




**NOTES**

1. Section at x=0, looking forward (+X). Local y=0 at track centre.
2. Two discs per station at y=+/-62, w40, D114.3, 1x 6205 each.
3. Disc bottom 34.05 on belt inner 34.04 (0.01 clearance).
4. Plate 6mm at y=+/-100; disc-to-plate gap 15.
5. Shaft D25 L260 clamped by collars at both plates; washer + R-pin.
6. No top roller: top run is unsupported (MBS: slack side).

**FLAIL MOWER 500 / UC-V2**

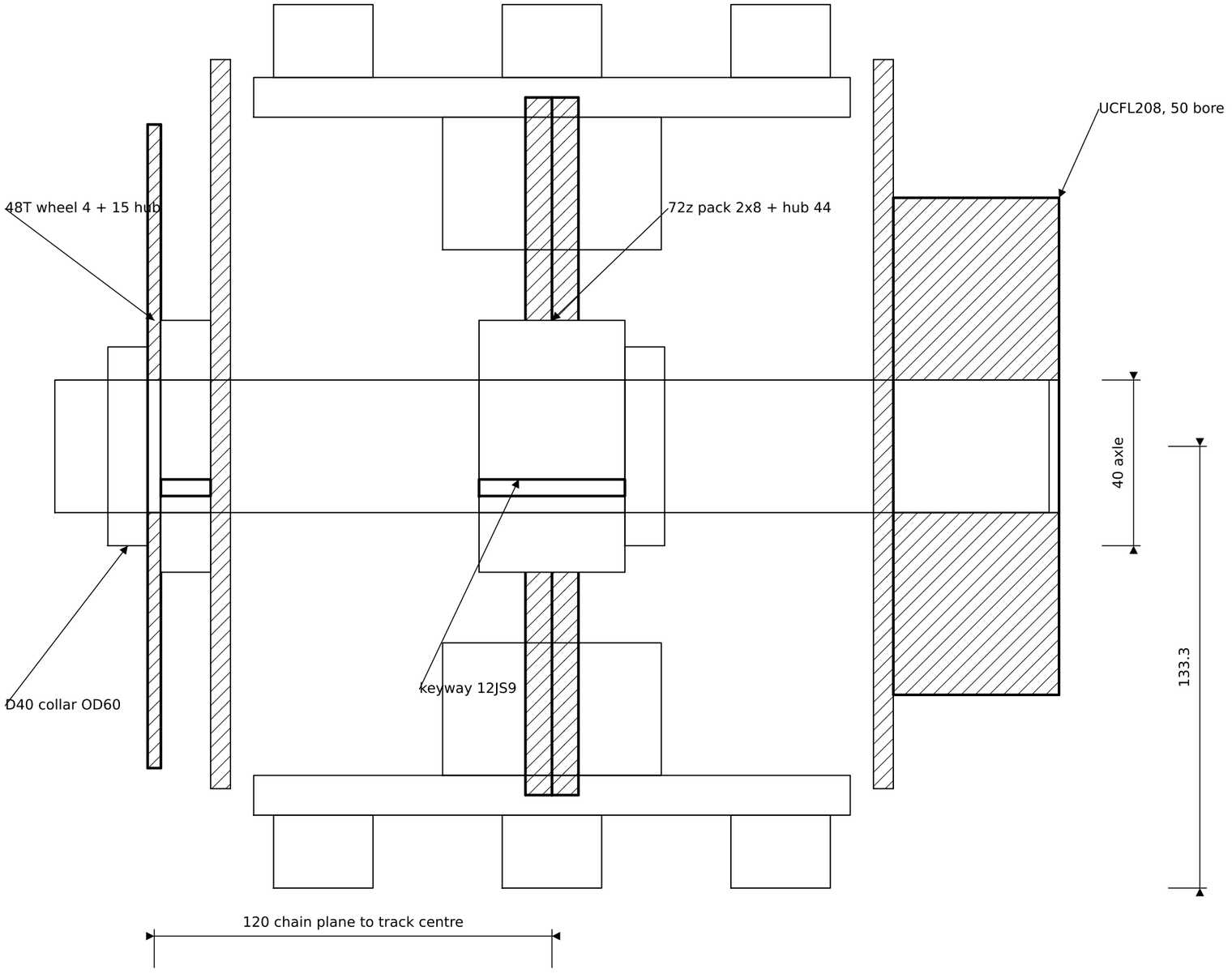
**SECTION A-A - ROLLER STATION**

DWG No.
**UCV2-GA-003**
REV A

| MATERIAL | QTY / MASS |
|---|---|
| IS2062 / EN8 | 1 station / - |

| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:1.5 | A3 | 3 / 14 | 2026-10-29 | OC |

## SECTION B-B (1:1.5)




**NOTES**

1. Section at x=-500.6, looking forward. Local y=0 at track centre.
2. 48T chain plane y=-120 (global 140); 72z pack at y=0 (global 260).
3. Axle D40 L300: keyways y=142..157 (48T) and 238..282 (72z hub).
4. UCFL208 outboard 363..413; axle end 410 -> 47mm engagement.
5. D40 collars (OD60) trap the 48T wheel and the 72z hub.

**FLAIL MOWER 500 / UC-V2**

**SECTION B-B - DRIVE STATION**

DWG No.

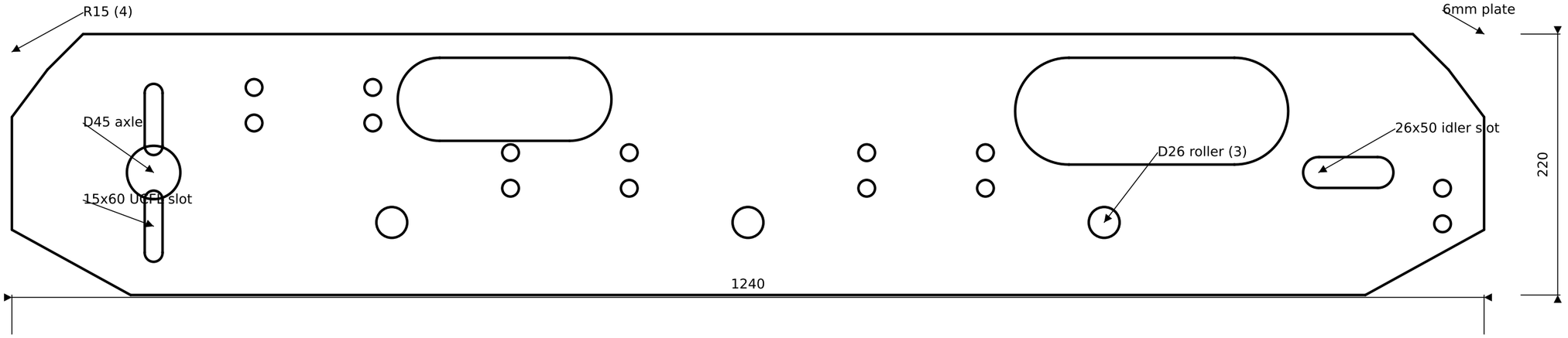
**UCV2-GA-004**

REV A

| MATERIAL | QTY / MASS |
|---|---|
| EN8 / IS2062 | 1 station / - |

| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:1.5 | A3 | 4 / 14 | 2026-10-29 | OC |

## FLAT PATTERN  (1:4)




DATUM: X0 = MID, Z0 = PLATE BOTTOM (z = local y + 30)

**HOLE TABLE (mm)**

| ID | x | z | size | qty |
|---|---|---|---|---|
| A | -300 / 0 / +300 | 91.2 | D26 | 3 |
| B | +480.6 | 133.3 | 26 x 50 | 1 |
| C | -500.6 | 133.3 | D45 | 1 |
| D | -500.6 | 88 / 178 | 15 x 60 | 2 |
| E | see P-007 | 120/150/175/205 | D14 | 14 |
| F | -280..-130 / 250..430 | 140..230 | 20R pocket | 2 |

**NOTES**

1. ONE DXF, mirror for L/R; etch part mark on non-show face.
2. Hole datum: X=0 at mid, Z=0 at plate bottom; table z is global.
3. Break all edges 1x45; corner radii R15 laser.
4. Plate seats: D26 roller (3), D45 axle, 26x50 idler slot, 15x60 UCFL slots.
5. 14x D14 holes take the beam flanges (schedule P-007).
6. 2 obround pockets 20R; no ribs (6mm dent-governed, see margins).

**FLAIL MOWER 500  /  UC-V2**

**SIDE PLATE - FLAT PATTERN**

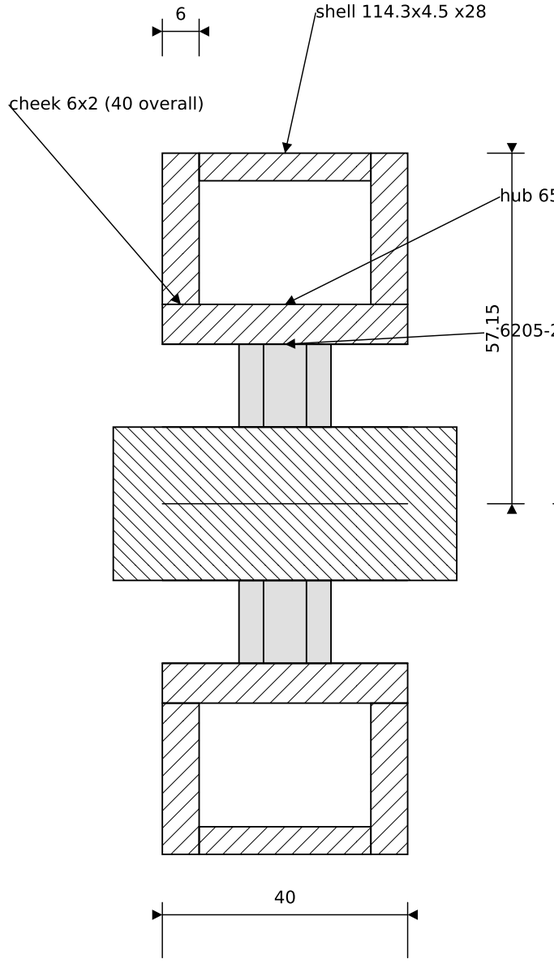
DWG No.

**UCV2-P-001**

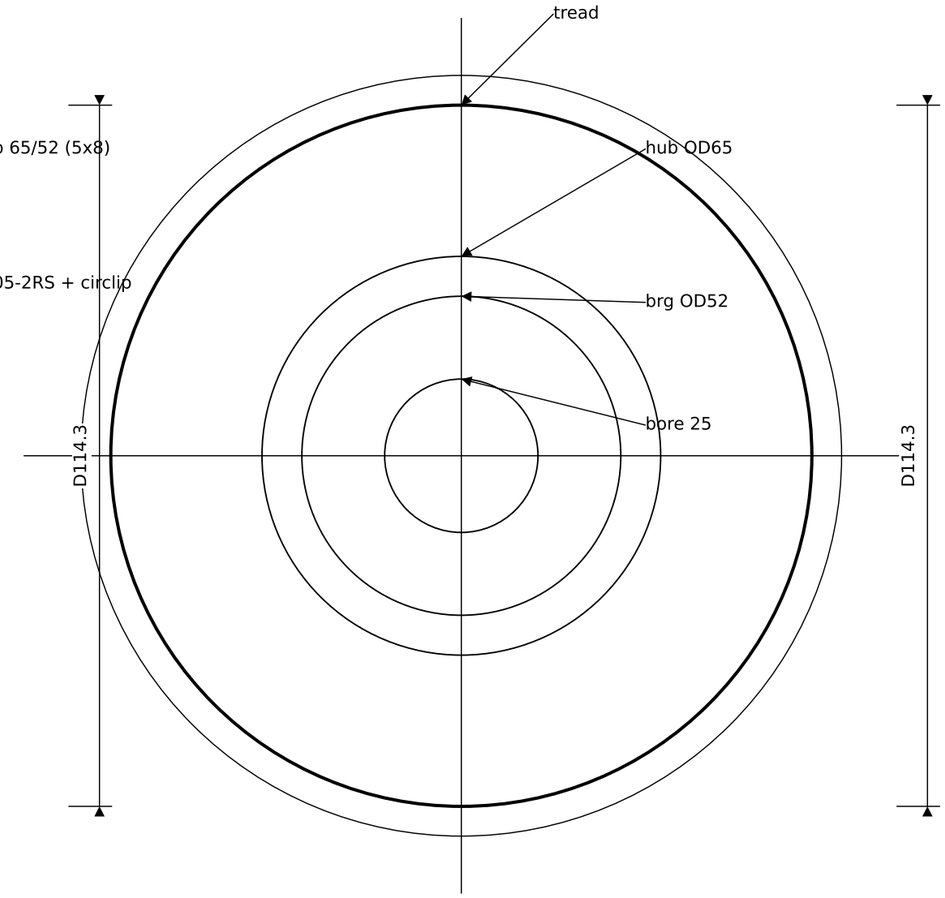
REV A

| MATERIAL | QTY / MASS | SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|---|---|
| IS2062 E250Br 6mm | 4 (2 L + 2 R)  /  10.27 kg ea | 1:4 | A3 | 5 / 14 | 2026-10-29 | OC |

## SECTION (1:1)




## END VIEW (1:1)




**DRUM PARTS**

| # | part | mat | qty | kg |
|---|---|---|---|---|
| 1 | shell 114.3x4.5x28 | MS-ERW | 1 | 0.34 |
| 2 | cheek OD114/ID65x6 | E250 | 2 | 0.32 |
| 3 | hub ring OD65/ID52x8 | E250 | 5 | 0.08 |
| 4 | 6205-2RS | COTS | 1 | 0.13 |
| 5 | circlip 52x1.5 | COTS | 1 | - |

**NOTES**

1. Shell 100NB-M 114.3x4.5 x40; cheeks 6mm OD114/ID65 (STRUCTURAL).
2. Hub: 5 laser rings OD65/ID52x8 = 40 long, Loctite 638 stack.
3. 1x 6205-2RS press + circlip 52x1.5. Sealed, no grease.
4. Weld cheeks OUTSIDE only, 3mm fillet, stitch then continuous.
5. Grind tread flush; no bead above OD114.3; runout <1.0.
6. Fit: hub OD into cheek ID65 with 0.05-0.15 Loctite gap.

**FLAIL MOWER 500 / UC-V2**

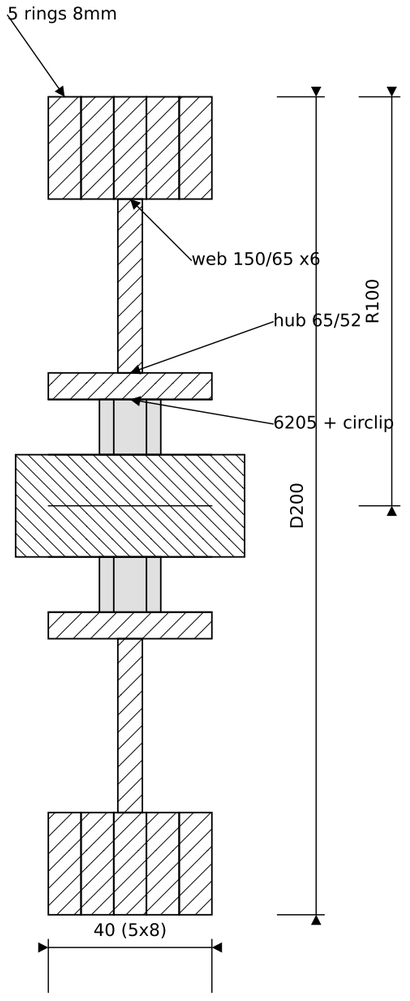
# ROLLER DRUM D114.3 - ASSEMBLY

DWG No.
**UCV2-P-002**
REV A

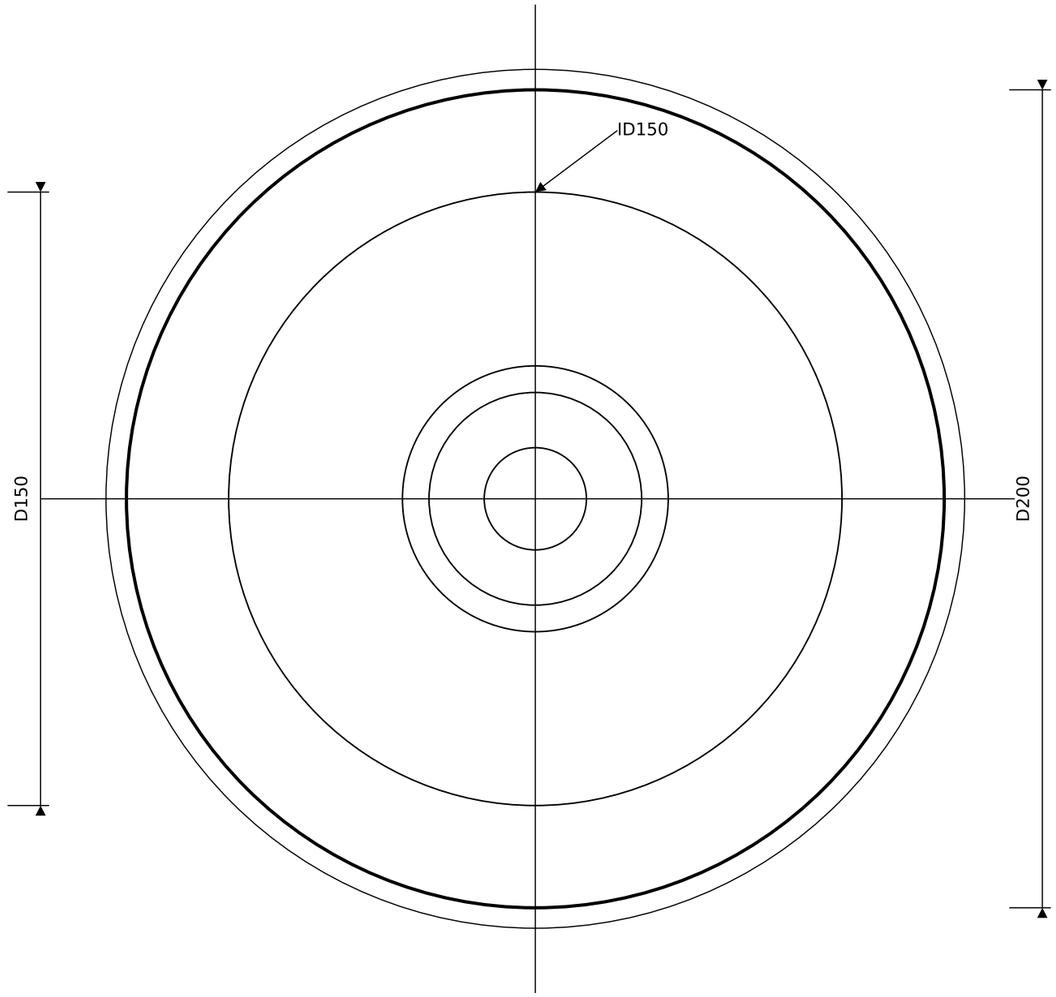
MATERIAL: MS-ERW + E250 + EN8
QTY / MASS: 12 / 1.37 + brg

SCALE: 1:1
SIZE: A3
SHEET: 6 / 14
DATE: 2026-10-29
DRAWN: OC

## SECTION (1:1.5)




## END VIEW (1:1.5)




**DISC PARTS**

| # | part | mat | qty | kg |
|---|---|---|---|---|
| 1 | ring OD200/ID150x8 | E250 | 5 | 0.86 |
| 2 | web OD150/ID65x6 | E250 | 1 | 0.68 |
| 3 | hub ring OD65/ID52x8 | E250 | 5 | 0.08 |
| 4 | 6205-2RS | COTS | 1 | 0.13 |
| 5 | circlip 52x1.5 | COTS | 1 | - |

**NOTES**

1. 5 laser rings OD200/ID150x8 = 40 wide tread, welded stack.
2. Web OD150/ID65x6 sits INSIDE the bore at mid-width (audit fix 2026-10-29).
3. Web is STRUCTURAL: connects tread to hub; weld both sides.
4. Hub: 5 rings OD65/ID52x8 = 40, Loctite 638.
5. 1x 6205-2RS press + circlip; no skim (flap-disc max 1mm).
6. Idler runs on belt inner R99.26; 0.7mm radial preload accepted.

**FLAIL MOWER 500 / UC-V2**

# IDLER DISC D200 - ASSEMBLY

DWG No.
**UCV2-P-003**
REV A

MATERIAL: E250 + EN8

QTY / MASS: 4 / 5.37 + brg

SCALE: 1:1.5

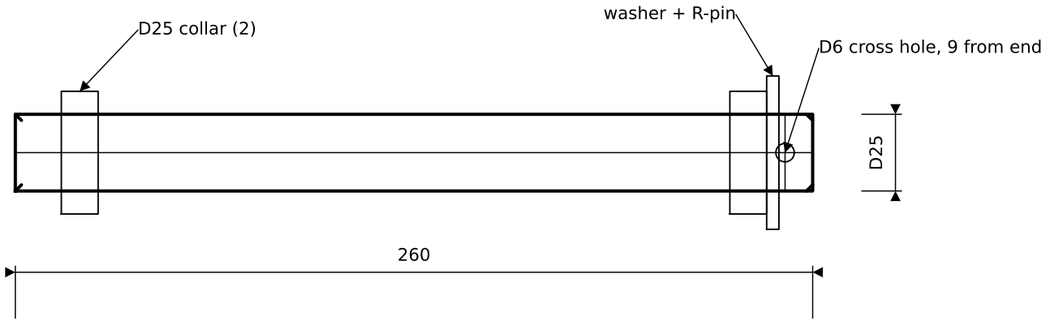
SIZE: A3

SHEET: 7 / 14

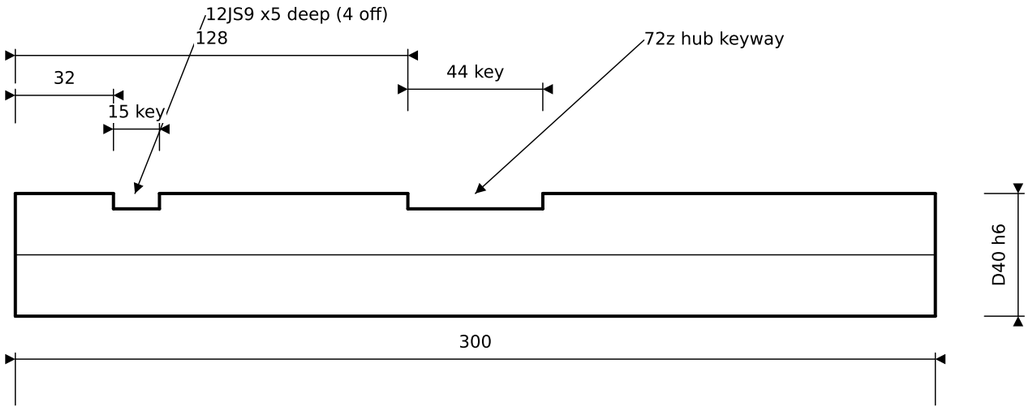
DATE: 2026-10-29

DRAWN: OC

## D25 SHAFT (1:2)




## D40 AXLE (1:2)




**NOTES**

1. D25 shafts: 8 off L260, EN8, g6; 1x D6 cross hole 9 from outboard end.
2. D40 axles: 2 off L300, EN8, h6; 4 keyways 12JS9 x5 deep (outsource).
3. Axle keyways: 32..47 and 128..172 from the 48T (inboard) end.
4. Collars/washer/R-pin shown dashed for position only (COTS).
5. Chamfer 1x45 both ends; protect keyways with tape for paint.

**FLAIL MOWER 500 / UC-V2**

# SHAFT D25 x260 & AXLE D40 x300

DWG No.

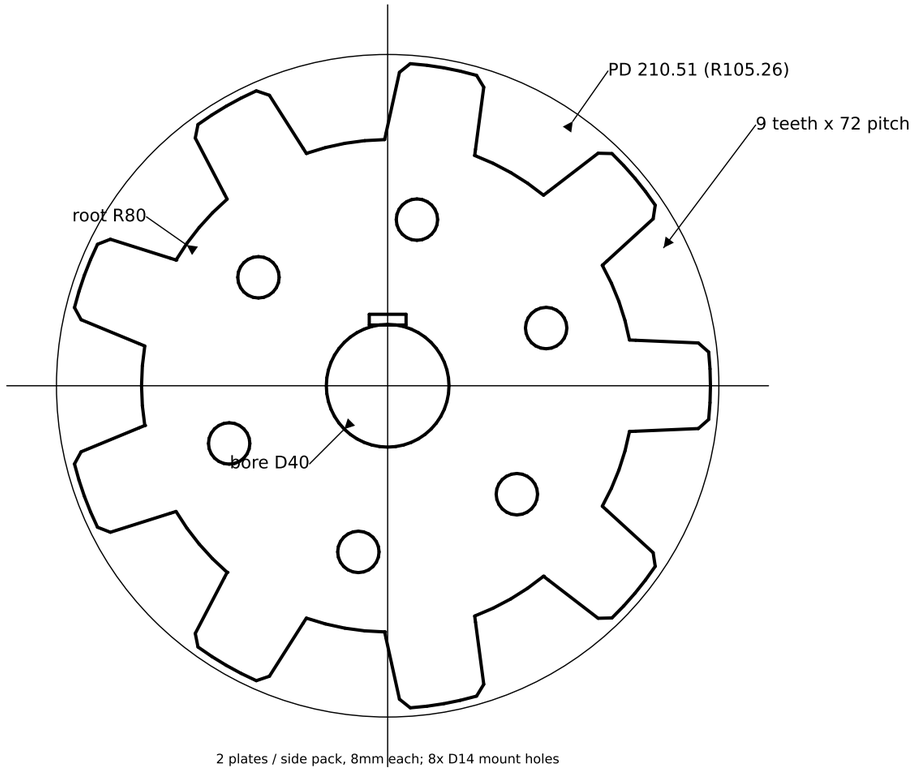
**UCV2-P-004**

REV A

| MATERIAL | QTY / MASS |
|---|---|
| EN8 (bright) | 8 + 2 / 1.00 / 2.93 kg |

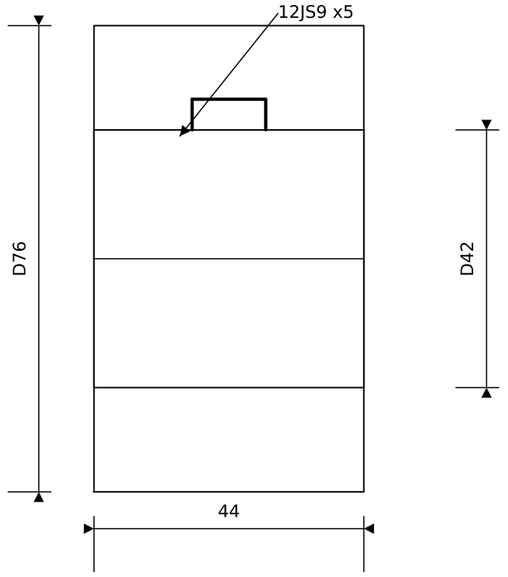
| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:2 | A3 | 8 / 14 | 2026-10-29 | OC |

## SPROCKET 72z9  (1:2)




2 plates / side pack, 8mm each; 8x D14 mount holes

## HUB  (1:1)




6x8 rings, faced

**DRIVE PARTS**

| part | description | mat | qty | kg |
|---|---|---|---|---|
| 72z9 | 9T x 72, PD210.51, D40, 8mm | E250 | 4 | 1.52 |
| hub76 | OD76/ID42 x44, 12JS9 | E250 | 2 | 1.13 |

**NOTES**

1. 72 mm pitch, 9 teeth, PD 210.51; laser 8mm E250, 2 plates/side pack.
2. Tooth profile per sprocket_72z9.dxf; tip R105.26, root R80.
3. Bore D40; hub carries the 12JS9 keyway (outsource).
4. Hub: 6 rings OD76/ID42x8 stacked 48, faced to 44; weld to pack.
5. Pack 2x8 = 16 wide, centred on track (y=+-260); hub 238..282.

**FLAIL MOWER 500  /  UC-V2**

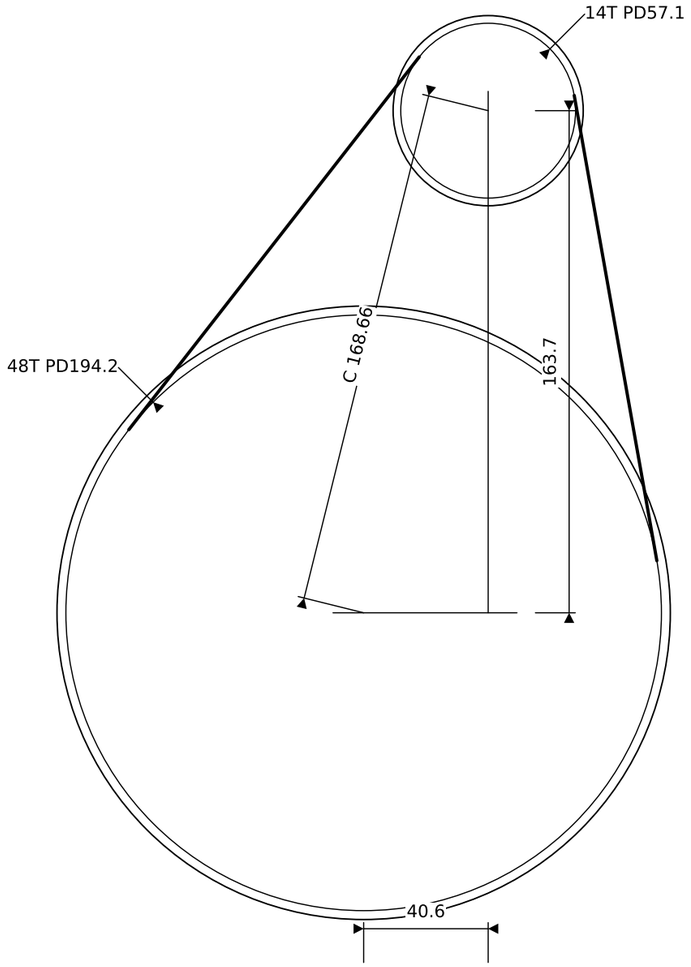
**SPROCKET 72z9  &  HUB 76/42**

DWG No.
**UCV2-P-005**
REV A

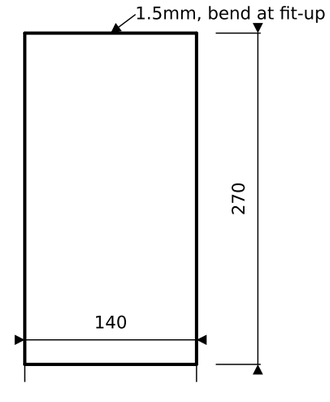
| MATERIAL | QTY / MASS |
|---|---|
| E250 8mm | 4 + 2  /  1.52 / 1.13 kg |

| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:2 / 1:1 | A3 | 9 / 14 | 2026-10-29 | OC |

## CHAIN GEOMETRY (1:2)




## GUARD (1:5)




### CHAIN DATA

| item | value | item | value |
|---|---|---|---|
| pitch | 12.7 (428) | links | 60 |
| 14T PD | 57.07 | 48T PD | 194.20 |
| ratio | 3.43 | pull | 835 N |
| C | 168.66 | sag | 10-15 |

**NOTES**

1. 428 pitch 12.7; 14T pinion -> 48T wheel, ratio 3.43.
2. 60 links at C~170 (motor rail slots take up 10-15mm sag).
3. Chain plane y=+-140; pinion and 48T wheel both on 140.
4. Chain pull 835N at 81Nm; SF 12 on a 20kN 428 chain.
5. Guard 1.5mm, 140x270 flat, bend at fit-up; vent slots.

**FLAIL MOWER 500 / UC-V2**

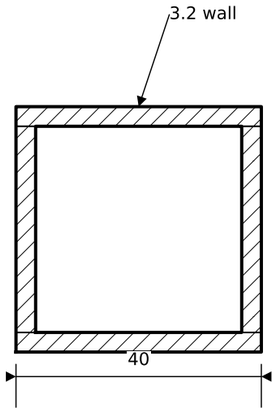
# CHAIN DRIVE 14T/48T & GUARD

DWG No.
**UCV2-P-006**
REV A

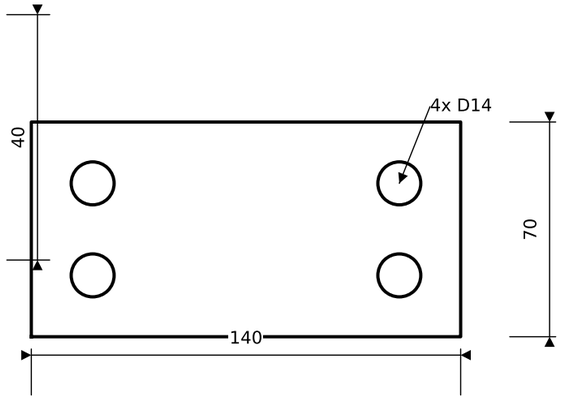
| MATERIAL | QTY / MASS |
|---|---|
| 428 chain / E250 guard | 2 sets / - |

| SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|
| 1:2 | A3 | 10 / 14 | 2026-10-29 | OC |

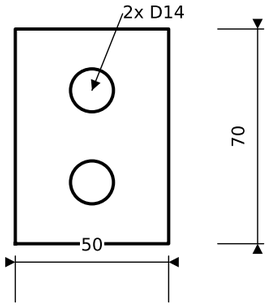
## SHS40 (1:1)



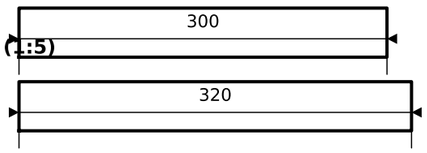

## FLANGE MID (1:2)




## FLANGE FRONT (1:2)




## BEAM L320 / RAIL L300 (1:5)




## BOLT SCHEDULE

| joint | flange | qty | M12 |
|---|---|---|---|
| mid | 140x70x8 | 4 | 16 |
| rear | 140x70x8 | 2 | 8 |
| front | 50x70x8 | 2 | 4 |
| total | - | 8 | 28 |

**NOTES**

1. All beams SHS40x40x3.2, one size; 4x L320 fitted (+1 spare).
2. Rails 4x L300 at z210..250; motors sit at z250.
3. Flange mid 140x70x8 (6 off) 4xM12; front 50x70x8 (2 off) 2xM12.
4. 28x M12 total per machine (14/side): mid 16, rear 8, front 4.
5. TVS 8.8, 80Nm, nylock + hardened washer + paint mark.
6. Weld 3mm fillet tube-to-flange, stitch then continuous.

**FLAIL MOWER 500 / UC-V2**

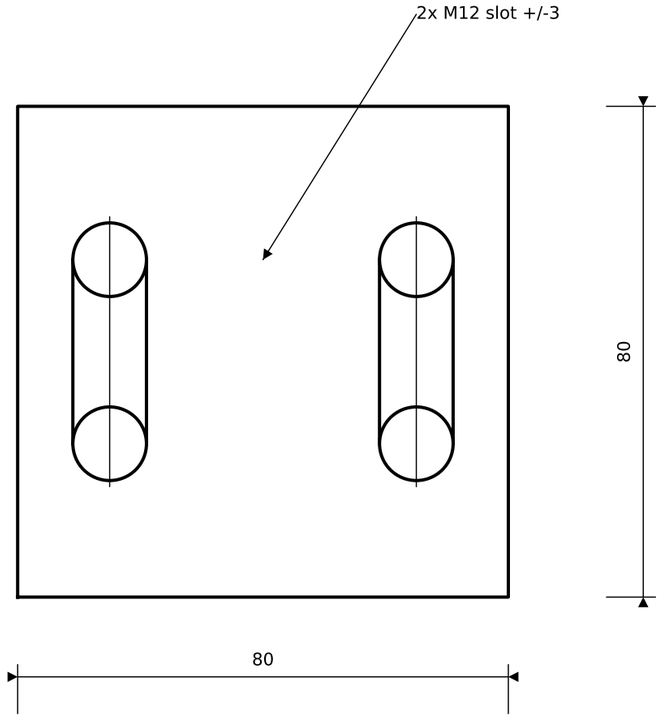
# FRAME SECTIONS, FLANGES & BOLT SCHEDULE

DWG No.
**UCV2-P-007**
REV A

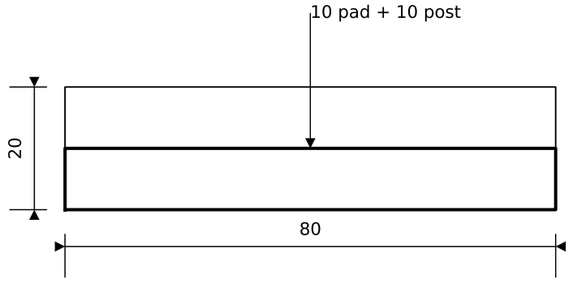
MATERIAL: SHS40x3.2 / E250 8mm
QTY / MASS: as listed / see table

SCALE: 1:1 / 1:2
SIZE: A3
SHEET: 11 / 14
DATE: 2026-10-29
DRAWN: OC

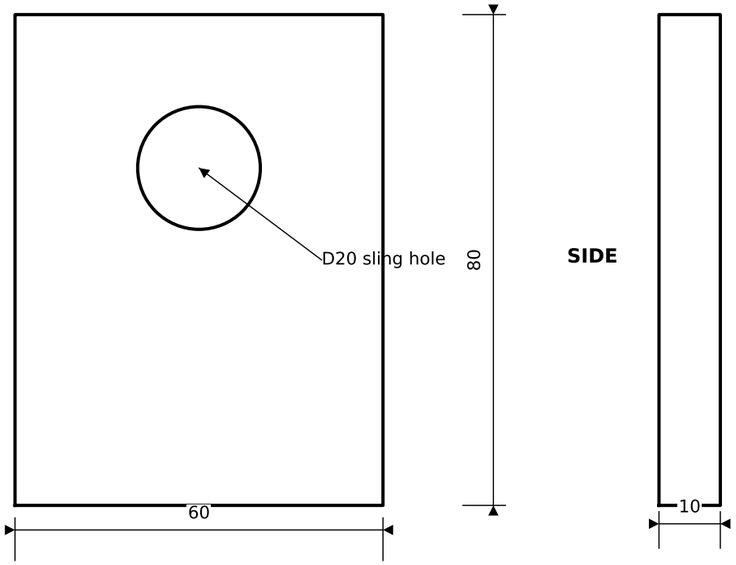
## DECK PAD (1:1)



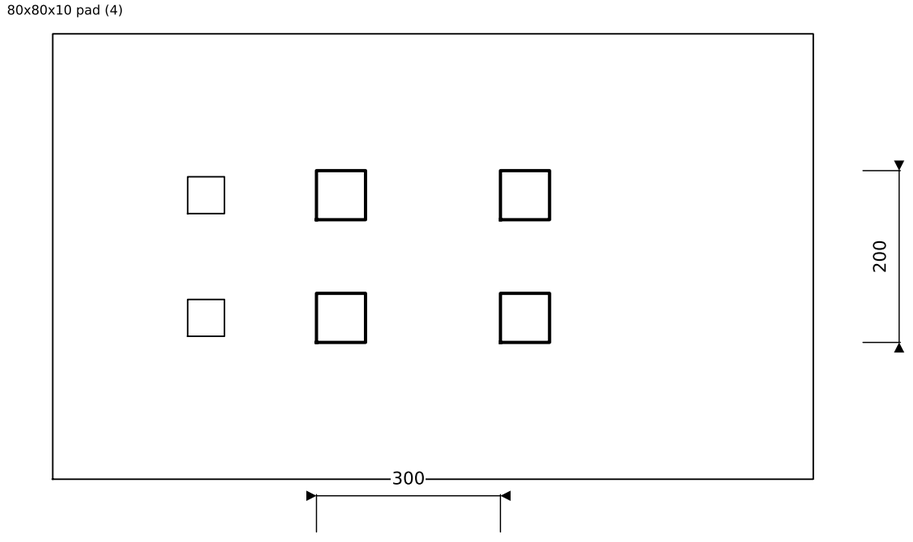

## SECTION (1:1)




## LIFT LUG (1:1)




## PAD / LUG POSITIONS (1:10)




### DECK PARTS

| part | size | mat | qty | kg |
|---|---|---|---|---|
| deck pad | 80x80x10 | E250 | 4 | 0.95 |
| lift lug | 10x60x80, D20 | MS | 2 | 0.35 |

**NOTES**

1. 4x deck pads 80x80x10 on mid beams, top z=175.
2. Pad centres x=+/-150, y=+/-100; M12 slot +/-3 + shim pack.
3. 2x lift lugs under the rear tie, x=-370, y=+/-100, z130..210.
4. Lug hole D20; sling rating to be marked at build.
5. Deck must clear motors (top z441) and overall W826.

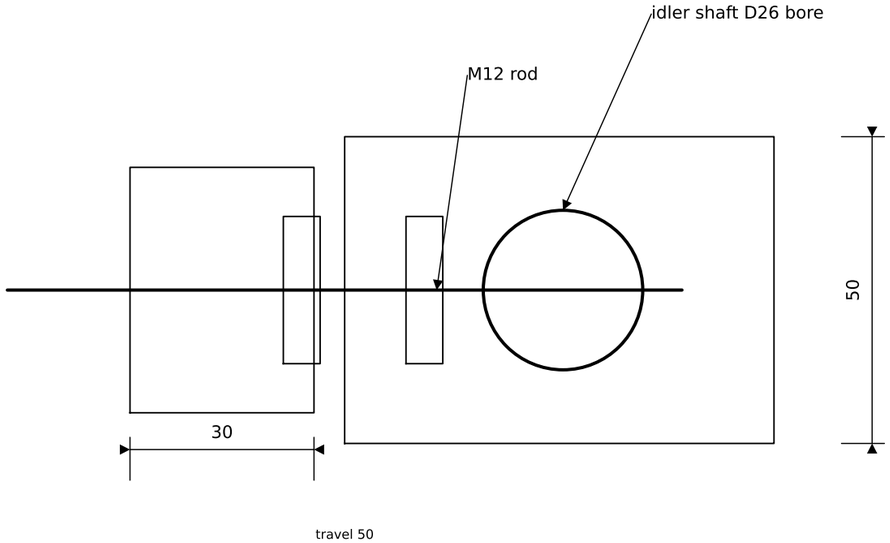
**FLAIL MOWER 500 / UC-V2**

# DECK INTERFACE & LIFT LUGS

DWG No. **UCV2-P-008** REV A

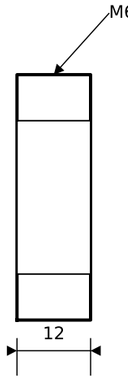
| MATERIAL | QTY / MASS | SCALE | SIZE | SHEET | DATE | DRAWN |
|---|---|---|---|---|---|---|
| E250 / MS | 4 pads + 2 lugs / see table | 1:1 / 1:10 | A3 | 12 / 14 | 2026-10-29 | OC |

## TENSIONER (1:1)

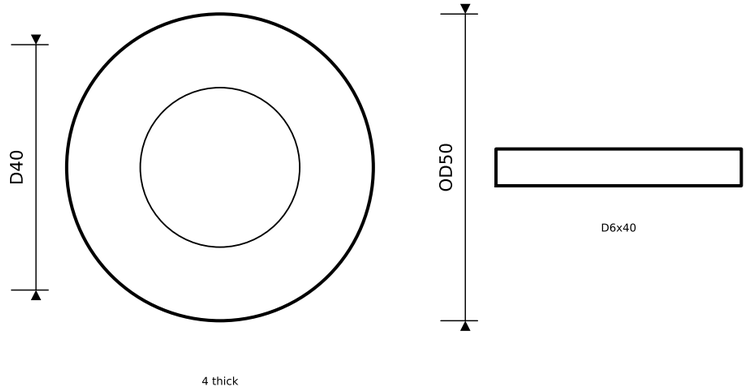



travel 50

## D25 COLLAR (1:1)




## WASHER / PIN (1:1)




4 thick

### SMALL PARTS

| part | size | mat | qty | kg |
|---|---|---|---|---|
| tension lug | 30x24x40 | MS | 4 | 0.20 |
| rod | M12x110 | EN8 | 4 | 0.10 |
| eye | 70x24x50 | MS | 4 | 0.52 |
| nut | M12 (19 A/F) | 8.8 | 8 | 0.02 |
| collar | D25 OD40x12 | COTS | 16 | 0.07 |
| collar | D40 OD60x12 | COTS | 4 | 0.15 |

**NOTES**

1. Idler slider travel 50; M12 rod pulls the eye/shaft.
2. Rod M12x110, eye 70x24x50, lug 30x24x40, nut M12 (19 A/F).
3. Set track sag 20 mid-span; jam nut after setting.
4. D25 collars x16 (OD40), D40 collars x4 (OD60), washers x8, pins x8.

**FLAIL MOWER 500 / UC-V2**

# TENSIONER & SMALL PARTS

DWG No.
**UCV2-P-009**
REV A

MATERIAL: EN8 / MS
QTY / MASS: 4 sets / see table
SCALE: 1:1
SIZE: A3
SHEET: 13 / 14
DATE: 2026-10-29
DRAWN: OC

## PARTS LIST (as drawn)

| cat | part | qty | kg ea | kg |
|---|---|---|---|---|
| structure | side plate 1240x220x6 | 4 | 10.274 | 41.1 |
| running | roller drum D114.3x40 | 12 | 1.37 | 16.44 |
| running | idler disc D200x40 | 4 | 5.367 | 21.47 |
| running | shaft D25x260 | 8 | 1.002 | 8.01 |
| running | collar D25 | 16 | 0.071 | 1.14 |
| running | washer D25/OD50 | 8 | 0.045 | 0.36 |
| running | pin D6x40 | 8 | 0.009 | 0.07 |
| drive | side axle D40x300 | 2 | 2.935 | 5.87 |
| drive | collar D40 OD60 | 4 | 0.146 | 0.58 |
| drive | sprocket 72z9 8mm | 4 | 1.521 | 6.08 |
| drive | hub 76/42 stack | 2 | 1.133 | 2.27 |
| structure | beam SHS40 L320 | 4 | 1.183 | 4.73 |
| structure | rail SHS40 L300 | 4 | 1.109 | 4.44 |
| structure | flange mid 140x70x8 | 6 | 0.577 | 3.46 |
| structure | flange front 50x70x8 | 2 | 0.2 | 0.4 |
| structure | deck mount 80x80x20 | 4 | 0.95 | 3.8 |
| structure | lift lug | 2 | 0.354 | 0.71 |
| drive | chain guard 1.5mm | 2 | 0.593 | 1.19 |
| drive | tension lug | 4 | 0.195 | 0.78 |
| drive | tension rod M12 | 4 | 0.098 | 0.39 |
| drive | rod eye | 4 | 0.52 | 2.08 |
| structure | spacer tube 48x3.2x200 | 4 | 0.71 | 2.84 |
| running | 6205-2RS | 16 | 0.13 | 2.08 |
| drive | 48T chainwheel (adapted) | 2 | 1.6 | 3.2 |
| drive | UCFL208 | 4 | 2.2 | 8.8 |
| drive | motor 1.2 kW + gearbox | 2 | 9.0 | 18.0 |
| drive | 428 chain (60L) | 2 | 0.8 | 1.6 |
| track | track 180x72x37 | 2 | 18.0 | 36.0 |
| misc | fasteners/paint/misc | 1 | 2.0 | 2.0 |

## MASS BY CATEGORY

| category | kg |
|---|---|
| structure | 61.4 |
| drive | 51.0 |
| running | 49.6 |
| track | 36.0 |
| misc | 2.0 |
| TOTAL | 199.9 |

**NOTES**

1. Mass measured from STEP at 7850 kg/m3; bought parts catalogue.
2. Total 199.9 kg incl tracks + motors (target 165 kg).
3. Laser parts: 10 DXFs (plate, sprocket, rings, web, cheeks, hubs, flanges, guard).
4. Outsource: 8 keyways + 2x 48T bore adapt; laser + machine shop only.
5. TVS 8.8 fasteners; 80Nm structural M12; nylock + paint mark.

**FLAIL MOWER 500 / UC-V2**

**PARTS LIST / BOM**

DWG No. **UCV2-P-010**

REV A

MATERIAL: as listed

QTY / MASS: 1 machine / 199.9 kg

SCALE: -

SIZE: A3

SHEET: 14 / 14

DATE: 2026-10-29

DRAWN: OC